

Youssuf Helaly

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EDUCATION

University of Ottawa

Bachelor of Applied Science in Software Engineering (Co-op) | GPA: 3.95/4.0

Expected Jun. 2027

Ottawa, ON

EXPERIENCE

AI/ML Engineering Consultant

May 2026 – Present

ThinkRF

Remote

- Architected a **multi-model ML platform** processing 50M+ real-time RF observations from 50+ global sensors; integrated **Isolation Forest** and **LSTM** models for large-scale spectrum anomaly detection and classification.
- Engineered a high-throughput **AWS data pipeline** ingesting 10M+ daily records; automated **CGI-based enrichment** to map global cellular identities across 5 countries in real-time.
- Developed an **AI Spectrum Intelligence engine** leveraging **Claude** with structured prompt chaining to transform raw RF telemetry into automated root-cause reports, replacing 3+ hours of daily manual analyst work.

Software Engineering Intern

Sep. 2025 – Dec. 2025

Trend Micro

Ottawa, ON

- Enhanced a Go-based **microservices platform** with REST APIs and Prometheus monitoring, deployed on **cloud infrastructure**, improving reliability and enabling 30% higher-scale simulations for developer workflows.
- Designed and built a **real-time data pipeline** ingesting, validating, and transforming **cybersecurity** logs into structured attack narratives with phase detection, processing 1000+ events per attack.
- Architected an **AI-driven environment generator** using **MCP** to create 100+ realistic environments with dynamic network topologies and embedded CVEs, reducing manual configuration time by 80%

Full-Stack Engineering Intern

Jan. 2025 – May 2025

Nokia

Ottawa, ON

- Improved system reliability by 20% by fixing service-layer defects and reducing Tomcat log errors in **Java Spring**.
- Automated **Elasticsearch** index config, cutting data inconsistencies by 40% and raising backend data quality.
- Refactored **React** frontend architecture with **Redux** state management and optimized **REST API** integration, cutting page load times by ~25% and improving form responsiveness across key user flows.

Software Engineering Intern

May 2024 – Aug. 2024

Solink

Ottawa, ON

- Engineered 20+ **Python RESTful API** endpoints via a **YAML-driven** config system and architected 50+ automated test cases, cutting deployment time by 25% and manual QA by 30%.
- Automated **CI/CD** test execution pipelines, eliminating \$2,000+/month in testing costs, compressing release cycles, and resolving flaky test root causes to improve build reliability.

PROJECTS

ClipIt (Proprietary) | TypeScript, Rust, TensorFlow, Firebase, React Native

- Built a cross-platform **React Native (iOS/Android)** real-time video intelligence app with a **3-tier ML pipeline** (YAMNet + contextual models), hitting <500ms latency and 85% detection accuracy at 30fps.
- Optimized on-device **ML** inference via **TensorFlow.js** + **WebGL** GPU acceleration and **TypedArrays**, sustaining 30+ second rolling buffers with zero memory leaks.
- Engineered a **Rust** backend (**Actix-Web** + **FFmpeg**) handling concurrent 50MB video uploads with sub-second processing and stitched output delivery under production load.

🔗 Image Analysis Web Application | ReactJS, Flask, AWS, SQL

- Developed a scalable cloud-based web app for image analysis using **React** and **Flask**, leveraging **AWS Rekognition** for image processing and **S3** for storage, improving search efficiency by 40%.
- Managed image uploads, user authentication, and filtering, processing 500+ images in seconds.
- Optimized **SQLite** database queries to enhance application performance and reduce processing redundancy.

TECHNICAL SKILLS

Languages: JavaScript, Python, Rust, Go, SQL, Java, TypeScript, HTML, CSS

Libraries/Frameworks: Spring Boot, React (Native), TensorFlow, Pandas, Scikit-learn, Flask, Redux, Next.js

Tools: Git, Docker, Kubernetes, AWS, Firebase, Linux, Elasticsearch, Claude, PostgreSQL, Redis, MCP, Prometheus